

ISE 555 : Deterministic Systems

Chapter 1 : Introduction to Operations Research

- World War II 'The art of winning the war without actually fighting it' - Arthur W. Clarke

• British - efficient ocean transport, effective bombing

• U.S - fighting patterns, planning, utilization of electronics

- 'The application of methods of science to complex problems in the direction and management of large systems of men, machines, materials and money in industry business and defence... Develop a scientific model of the system, incorporating measurements of factors such as chance and risk with which to predict and compare the outcomes of alt. decisions, strategies or controls. Help management determine its policy and actions scientifically.'

- Applications

• Budgeting & Investments

• Marketing

• R&D

• Purchasing

• Personal Management

• Production Management

- Phases

1. Formulation of the problem - objectives, variables, ranges
2. Construction of a mathematical model - iconic, analogue, ^{symbolic} mathematical
3. Deriving a solution from the model - optimize for objection ^{+ sensitivity}
4. Testing model and solution - solution of model should perform best
5. Implementing a solution - Be ready solution can get outdated

- Scope - Wide variety of industry applies OR any strategy

Modelling in OR - 'an idealized representation of a real-life system'

- Advantage - Concise, logical, systematic, enables high powered math
- Disadvantage - Not to be considered absolute, verified through experiment, ^{not} timely

* Iconic - a pictorial representation of a system, shrunk up or scaled down for purpose. Globe, atom diagram, map. Difficult to manipulate for experiment

* Analogue - one set of properties used to represent another, graphs distance $\rightarrow$ variable, pi-graph area $\rightarrow$ percentage/probability. Not very specific

* Mathematical (Symbolic) Model - utilizes equations to predict behaviors

(i) - deterministic : parameter assumed known & precise (specific)

(ii) nondeterministic : at least one parameter is random or fuzzy (general)

~~Specific time = Specific, Time considered = General + Dynamic~~

~~General Solutions~~

Specific or ~~General~~ (time factor)

Static or Dynamic

General (non-optimized)

General Solutions

- Analytical Methods - basic math, differentials, calculus
- Numerical Methods - trial & error, machine learning, continuous improvement
- Monte Carlo Technique - Random number table, plug and find

Jean Notes:

◦ Elementary calculus is not always enough, several turning points.
local vs. global min/max, endpoint checking

◦ Standard problem-formulation

1. Decision & scope
2. Given Data
3. Decision Variables
4. Objective
5. Constraints
6. Domain & assumptions
7. Validate and interpret

4. Two-variable linear program

	Coords	Objective Value
$\max f(x) = 2x_1 + x_2$	A (1, 0)	2
subject to $x_1 + x_2 \leq 1$,	B (0, 1)	1
$x_1, x_2 \geq 0$	C (0, 0)	0

5. Cabinet Manufacturing

6. Insurance Workload allocation

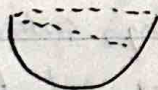
~~Non-negativity~~

Chapter 2: Convex and Concave Functions

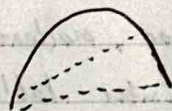
- Properties which can be used to develop suitable optimality conditions and computational schemes for optimization problems.

A function $f: S \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ (where S is a non-empty convex set) is convex at $\bar{x} \in S$, if

Convex



Concave



Convex - graph lies at or below every chord connecting two of its points

Concave - lies at or above

• Straight connecting segment a chord. The relationship must hold for every pair of points in the domain being studied.

$\lambda x_1 + (1-\lambda)x_2$ a weighted average of two inputs x_1 & x_2 . With $0 \leq \lambda \leq 1$ it selects an input between them. $\lambda = 0.5$ selects the midpoint

* Central Inequality

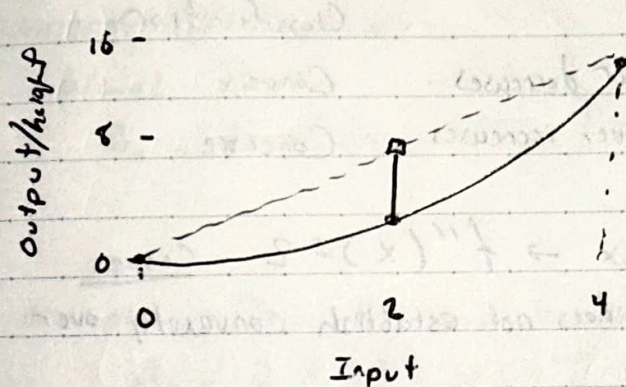
$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

border $\uparrow$ height of curve $\uparrow$ height of chord $\uparrow$ dashed line.
(mix inputs then apply) (calculate endpoints then mix outputs)

For $f(x) = x^2$ $x_1 = 0$, $x_2 = 4$ & $\lambda = 0.5$

Curve - Halfway input 2, square it = 4

Chord - Endpoints 0 & 16 (4^2) average = 8



Input	2
curve	4
Chord	8

- Apply the weighted function to a weighted average of inputs
- Take the weighted average of the endpoint outputs

Convexity compares two operations

• For a convex function, the first result $f(\text{input})$ is less than the second (output $\neq$ average) for concave it's $\leq$.

Set is a collection of allowed inputs

A set is convex if the entire straight segment between any two points is allowed. - must be continuous

Domain: the set of allowed inputs

Convex set: contains a continuous straight segment between any two points

Convex set describes allowed inputs

Convex or Concave function describes the relationship between a graph and cards.